How-To Library

Informatica

Configuring Ports for Big Data Management, Data Integration Hub, Enterprise Information Catalog, and Intelligent Data Lake

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Abstract

You can deploy a solution consisting of several Informatica products to address your requirements to extract, process and report data and metadata from big data sources. To prevent conflicts between products, this article tells you which ports are established when you run the installer for each product.

Supported Versions

- Informatica Intelligent Data Lake 10.1.1 Update 2
- Informatica Enterprise Information Catalog 10.1.1 Update 2
- Informatica Data Integration Hub 10.1.1 Update 2
- Informatica Big Data Management 10.1.1 Update 2

Table of Contents

Overview. . . . 2
Informatica Domain and Application Services. . . . 2
Application Services and Ports. . . . 3
Data Integration Hub Port Numbers 5
Live Data Map Ports. . . . 5
Enterprise Information Catalog Ports. . . . 7
Big Data Management Ports. . . . 7
IBM BigInsights Port Reference. . . . 7
Cloudera Port Reference. . . . 8
HDInsight Port Reference. . . . 9
Hortonworks Port Reference. . . . 10
MapR Port Reference. . . . 11
Hadoop Cluster Port Reference. . . . 11
Blaze Engine Port Reference 12
Developer Tool Port Reference. . . . 12
Intelligent Data Lake Ports. . . . 12

Overview

You can enable several Informatica products in a solution that works together to extract, process and report data and metadata.

When you install Informatica products, the installer utility opens specific ports by default. You must open other ports manually. This article tells you which ports are established when you run Informatica product installers.

Informatica Domain and Application Services

The Informatica domain includes several services that perform important roles in data extraction and processing.

For more information about application services, see the *Informatica 10.1 Application Service Guide*.

Application Services and Ports

Informatica domain services and application services in the Informatica domain have unique ports.

Informatica Domain

The following table describes the ports that you can set:

Port	Description
Service Manager port	Port number used by the Service Manager on the node. The Service Manager listens for incoming connection requests on this port. Client applications use this port to communicate with the services in the domain. The Informatica command line programs use this port to communicate to the domain. This is also the port for the SQL data service JDBC/ODBC driver. Default is 6006.
Service Manager Shutdown port	Port number that controls server shutdown for the domain Service Manager. The Service Manager listens for shutdown commands on this port. Default is 6007.
Informatica Administrator port	Port number used by Informatica Administrator. Default is 6008.
Informatica Administrator HTTPS port	No default port. Enter the required port number when you create the service. Setting this port to 0 disables an HTTPS connection to the Administrator tool.
Informatica Administrator shutdown port	Port number that controls server shutdown for Informatica Administrator. Informatica Administrator listens for shutdown commands on this port. Default is 6009.
Minimum port number	Lowest port number in the range of dynamic port numbers that can be assigned to the application service processes that run on this node. Default is 6014.
Maximum port number	Highest port number in the range of dynamic port numbers that can be assigned to the application service processes that run on this node. Default is 6114.

Analyst Service

The following table lists the default port associated with the Analyst Service:

Type	Default Port
Analyst Service (HTTP)	8085
Analyst Service (HTTPS)	No default port. Enter the required port number when you create the service.
Analyst Service (Staging database)	No default port. Enter the database port number.

Content Management Service

The following table lists the default port associated with the Content Management Service:

Type	Default Port
Content Management Service (HTTP)	8105
Content Management Service (HTTPS)	No default port. Enter the required port number when you create the service.

Data Director Service

The following table lists the default port associated with the Data Director Service:

Type	Default Port
Data Director Service (HTTP)	No default port. Enter the required port number when you create the service.
Data Director Service (HTTPS)	No default port. Enter the required port number when you create the service.

Data Integration Service

The following table lists the default port associated with the Data Integration Service:

Type	Default Port
Data Integration Service (HTTP proxy)	8080
Data Integration Service (HTTP)	8095
Data Integration Service (HTTPS)	No default port. Enter the required port number when you create the service.
Profiling Warehouse database	No default port. Enter the database port number.
Human Task database	No default port. Enter the database port number.

Metadata Manager Service

The following table lists the default port associated with the Metadata Manager Service:

Type	Default Port
Metadata Manager Service (HTTP)	Default is 10250.
Metadata Manager Service (HTTPS)	No default port. Enter the required port number when you create the service.

PowerExchange® Listener Service

Use the same port number that you specify in the SVCNODE statement of the DBMOVER file.

If you define more than one Listener Service to run on a node, you must define a unique SVCNODE port number for each service.

PowerExchange Logger Service

Use the same port number that you specify in the SVCNODE statement of the DBMOVER file.

If you define more than one Listener Service to run on a node, you must define a unique SVCNODE port number for each service.

Web Services Hub Service

The following table lists the default port associated with the Web Services Hub Service:

Type	Default Port
Web Services Hub Service (HTTP)	7333
Web Services Hub Service (HTTPS)	7343

Data Integration Hub Port Numbers

The Data Integration Hub installer sets the default port numbers for the installation components. If another application uses the same port number as one of the installation components, a port conflict might prevent the component from running correctly or cause errors.

You can change the port numbers after installation. Before you start Data Integration Hub, verify that the port numbers do not conflict with other applications and change the port numbers in Data Integration Hub to prevent port conflicts.

The following table describes the default port numbers:

Description	Port Number
UDP multicast port that Data Integration Hub uses for internal communications.	18000
Operation Console shutdown port. Only required to be available on the machine where Data Integration Hub is installed.	18005
Port that the Operation Console uses for internal communications.	18050
Operation Console HTTP port. Required only if you use an HTTP port for the Operation Console.	18080
RMI port for Data Integration Hub server startup and shutdown.	18095
RMI ports that the Operation Console and PowerCenter workflows use to communicate with the Data Integration Hub server.	18095 and 10896
Port that the Data Integration Hub server uses for internal communications.	18100
Operation Console HTTPS port. Required only if you use an HTTPS port for the Operation Console.	18443
Port for the Data Integration Hub JNDI provider. This port is also the JMS listener port for the Data Integration Hub JMS Broker.	18616

Live Data Map Ports

The following table lists ports that the installer opens when you install Live Data Map:

Description	Port Number
Ambari Agent TCP	8670
Ambari Metrics Collector TCP	6188, 60200

Description	Port Number
Ambari Server TCP	8080, 8440, 8441
Ambari Server UDP	59804
HDFS DataNode	50010
HDFS DataNode HTTP	50075
HDFS DataNode HTTPS	50475
HDFS DataNode IPC	8010
HDFS HTTPS	50470
HDFS JournalNode HTTP	8480
HDFS JournalNode HTTPS	8481
HDFS NameNode HTTP	50070
HDFS NameNode HTTPS	50470
HDFS NameNode RPC	8020
HDFS Secondary Name Node	50090
pgSQL	5432
YARN JobHistory Server	19888
YARN NodeManager	45454
YARN ResourceManager	8032
YARN ResourceManager Admin	8141
YARN ResourceManager ResourceTracker	8025
YARN ResourceManager Scheduler	8030
YARN ResourceManager Webapp	8088
YARN ResourceManager Webapp HTTPS	8090
YARN Timeline Server	10020
YARN Timeline Server Webapp	8188
YARN Timeline Server Webapp HTTPS	8190
Zookeeper	2181

Enterprise Information Catalog Ports

The following table lists the ports that Enterprise Information Catalog uses:

Description	Port
Catalog Service	9085
Informatica Cluster Service	9075

Big Data Management Ports

The following sections list Hadoop components and default port numbers for various Hadoop distributions:

IBM BigInsights Port Reference

The following table lists the default ports associated with IBM BigInsights:

Hadoop Component	Port
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9933
HiveServer	10000
JobTracker	9001
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	9000
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030

Hadoop Component	Port
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

Cloudera Port Reference

The following table lists the ports associated with Cloudera:

Hadoop Component	Port
Cloudera Manager	7180
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9083
HiveServer	10000
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	8020
ResourceManager	8050
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562

Hadoop Component	Port
TaskTracker web management	50060
ZooKeeper	2181

HDInsight Port Reference

The following table lists the ports associated with HDInsight:

Component	Port
Namenode API port	8020
Job Tracker	8012, 8021, 9001
Resource Manager	8032, 8088
Scheduler Address	8030
Resource Tracker	8031
MapReduce JobHistory server	10020
HDFS read/write	50010, 50020
Hive Server	10000
Hive Metastore	9083, 9933
JobTracker web management	50030
TaskTracker web management	50060
Namenode web management	50070
Datanode web management	50075
Secondary namenode web management	50090
Backup/Checkpoint node web management	50105
Zookeeper port	21, 181, 815
mapreduce.shuffle.port	13562
yarn.app.mapreduce.am. job.client.port-range	50100 - 50200
Hadoop cluster ports	22, 1024, 1433, 1521, 2888, 3306, 3389, 3888, 4867, 5000-5003, 5678, 5678, 6262, 7180, 7180-7187, 7180-7187, 7183, 7184, 7222, 7432, 8010, 8025, 8050, 8080, 8083, 8083-8087, 8084, 8084, 8086, 8087, 8091, 8141, 8188, 8190, 8440, 8441, 8480, 8651, 8660, 8661, 8662, 8663, 9000, 9000, 9994-9997, 9994-9999, 10001, 10101, 10101, 10200, 18000-18025, 19001, 19001, 19888, 45454, 50000-50003, 50111, 50470, 60000, 60010, 60020, 60030,

Component	Port
RDP	3389
SSH & SFTP	22
FTP	21

Hortonworks Port Reference

The following table lists the ports associated with Hortonworks:

Hadoop Component	Port
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
Hive metastore	9933
HiveServer	10000
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
MySQL	3306
NameNode	8020
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

MapR Port Reference

The following table lists the ports associated with MapR:

Hadoop Component	Port	Notes
CLDB	7222	
CLDB JMX monitor	7220	
CLDB web port	7221	
HBase master	16000	
HBase master web	16010	
HBase region server	16020	
Hive metastore	9083	
HiveServer	10000	
JobTracker	9001	
JobTracker web manager	50030	
MapReduce Application Master	50100-50200	
MapReduce JobHistory server	10020	YARN only
MapReduce JobHistory server web app	19888	YARN only
MySQL	3306	Required if you use MySQL.
ResourceManager	8032	YARN only
ResourceManager web app	8088	YARN only
Secure Resource Manager web app (HTTPS)	8090	
ResourceTracker	8031	YARN only
Scheduler address	8030	YARN only
Shuffle HTTP	13562	
TaskTracker web management	50060	
ZooKeeper	5181	

Hadoop Cluster Port Reference

A Hadoop cluster requires some unique port numbers.

Blaze Engine Port Reference

The Blaze engine uses ports for the Blaze Job Monitor and the Blaze services.

Blaze Job Monitor

The following table lists the ports for the Blaze Job Monitor:

Port Name	Default Port Number
HTTP	9080
JSF	9090

Blaze Services

The following table lists the ports for Blaze services:

Port Name	Default Port Number
Minimum and Maximum Port Range for Hadoop Connection	12300 - 12600

Developer Tool Port Reference

The following table lists the ports for the Developer tool:

Port Name	Default Port Number
CLDB	7222 - MapR only
HBase master	60000
HBase region server	60200
HiveServer2 thrift	10000
NameNode RPC	8020 - except MapR
ZooKeeper client	2181, 5181

Intelligent Data Lake Ports

When you install Intelligent Data Lake, the installer opens the following ports.

Data Preparation Service

The following table lists the ports you can configure for the Data Preparation Service:

Description	Port Number
HTTP	8099
Solr	8983

Intelligent Data Lake Service

The following table lists the port you can configure for the Intelligent Data Lake Service:

Description	Port Number
HTTP	9045

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